

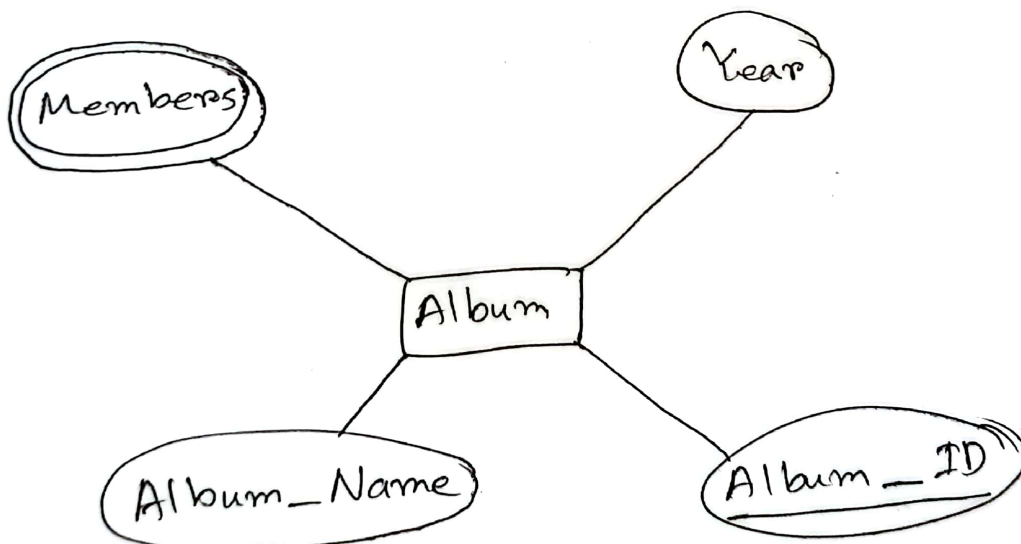
Full Name (in Block Letter): _____ ID: _____

Question 1 [CO2] : 3 Points

Represent the following information using ER Diagram symbols and identify different types of attributes used:

Album			
Album_Name	Album_ID	Year	Members
Closure/Continuation	A_2022_001	2022	Steven Wilson, Richard Barbieri, Gavin Harrison
The Future Bites	A_2021_001	2021	Steven Wilson, David Kosten, Michael Spearman

Question 1 Answer:

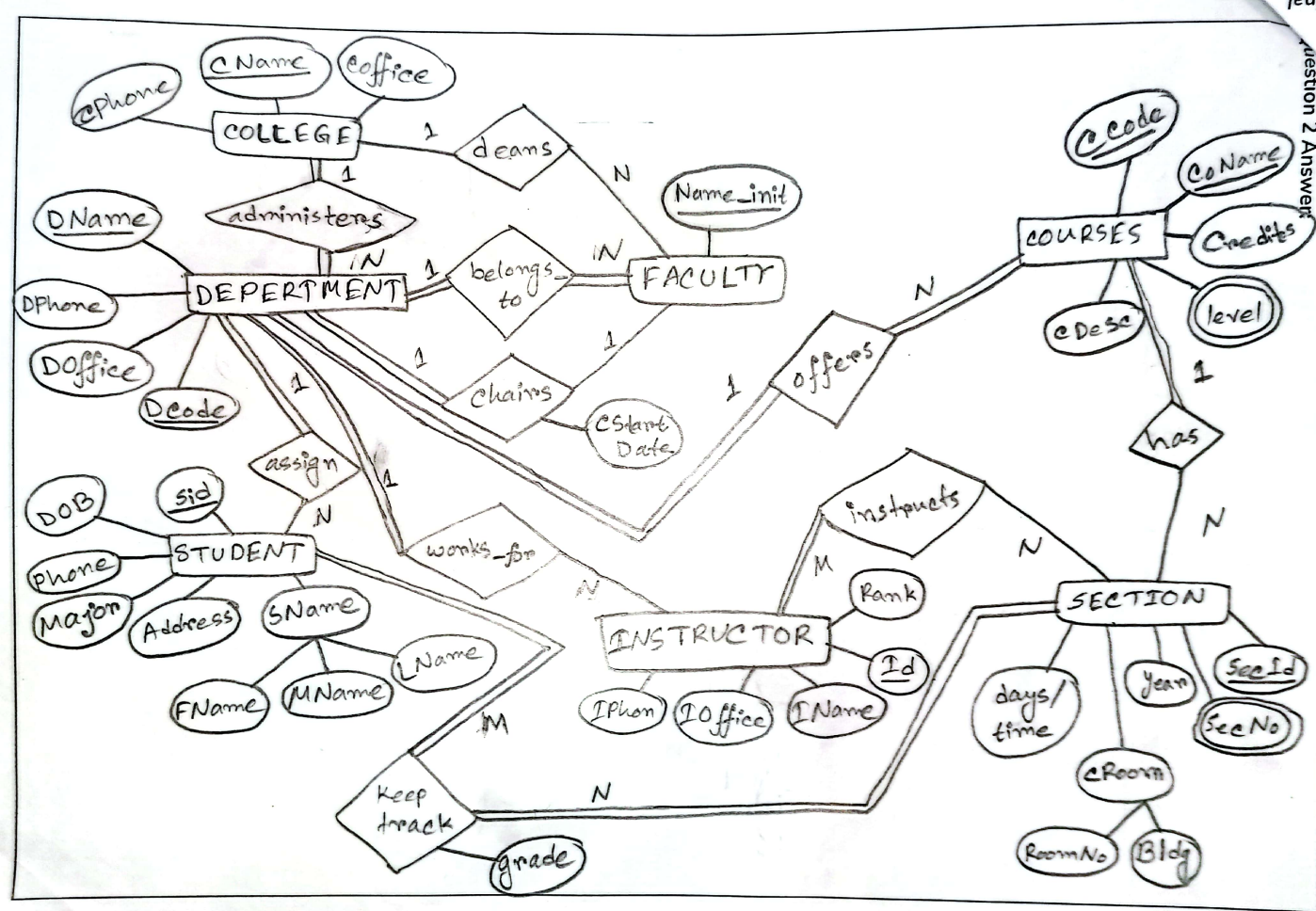


here,

Entity → Album

Attributes → Year, Album_ID, Album_Name, Members

Multivalued → Members; Key-Attribute → Album_ID



Question 3 [CO2] : 7 Points

Draw an ER diagram for keeping track of information about votes taken in the U.S. House of Representatives during the current two-year congressional session. The database needs to keep track of each U.S. STATE's Name (e.g., 'Texas', 'New York', 'California') and include the Region of the state (whose domain is {'Northeast', 'Midwest', 'Southeast', 'Southwest', 'West'}). Each CONGRESS_PERSON in the House of Representatives is described by his or her Name, plus the District represented, the Start_date when the congressperson was first elected, and the political Party to which he or she belongs (whose domain is {'Republican', 'Democrat', 'Independent', 'Other'}). The database keeps track of each BILL (i.e., proposed law), including the Bill_name, the Date_of_vote on the bill, whether the bill Passed_or_failed (whose domain is {'Yes', 'No'}), and the Sponsor (the congressperson(s) who sponsored—that is, proposed—the bill). The database also keeps track of how each congressperson voted on each bill (domain of Vote attribute is {'Yes', 'No', 'Abstain', 'Absent'}).

Use regular notation to denote relationship structural constraints (Cardinality Ratio and Existence Dependency Constraint). State clearly any assumptions you make.

Question 3 Answer:

